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ML Exercises, week 4.

Exercises:

Exercise 1 - generating test sets.

Read the following article/page to get some idea of what test data is easily available in Python with Sci-kit Learn:
https://scikit-learn.org/stable/datasets/toy_dataset.html

https://python-course.eu/machine-learning/available-data-sets-in-sklearn.php

Or see these pages using the wayback archive:
https://machinelearningmastery.com/a-guide-to-getting-datasets-for-machine-learning-in-python/

https://machinelearningmastery.com/generate-test-datasets-python-scikit-learn/

Exercise 2 - clustering.

We will now look at the "Moons Classification Problem". Here we look at a 2D dataset, that is labelled into to different classes (binary classification).
For this exercise we are not concerned about the labels, only the 2D data.
Try to use that data and first visualize the data - you can get some help from the code on the page. Note: we want more samples than 100, so specify the sample size to 1000. You can see the full documentation of how to generate the moon dataset here:
https://scikit-learn.org/stable/modules/generated/sklearn.datasets.make_moons.html

The next step is to try the K-means classification algorithm (for clustering) on this data and see what it comes up with (set the K-value to 2).
Try to visualize the resulting two clusters with colors.
How does the clustering compare the real categories?
(it will probably be quite different clustering shapes than the real shapes).
But this will show some of the limitations of standard clustering.

Exercise 3 - classification using a Decision Tree.

3.1 Split the moon dataset 80/20 (how to do this in python is part of this exercise) into a training set and test set (so the first 800 samples and their labels will be used for training, and the last 200 samples will be used to test how good the training was).
3.2 Train a Decision Tree on the first 800 samples.
3.3 Then test the decision tree on the last 200 samples and calculate the percentage it classifies correctly. A completely random classifier would be expecting to get 50% correct, hopefully you will do better!
3.4 Try to visualize the resulting decision tree.

3.5. Find examples it classifies correctly, and incorrectly.

Exercise 4 - classification using a Random Forest

Do the same as in exercise 3 (Questions 1,2 & 3), but now instead of using a single tree then use a Random Forest.
You can try to experiment a bit with the number of trees in the forest.
Finally, visualize how more trees improves the overall result (See the RandomForest tutorial python file on Canvas for inspiration).

Exercise 5 - classification on other data.

Try exercise 3 but on the Circles dataset instead (must be labeled data of course) from the article in exercise 1.

Exercise 6 - hard and soft voting.

What is the difference between hard and soft voting classifiers?
Write it down in your own words.